WASP-36b

R-0668

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_260228` · created 2026-07-27T18:17:10+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461099.7529 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.045
Transit depth (Rp/Rs)^2	0.88 +/- 0.83 [%]
Orbital Inclination (inc)	87.93 +/- 2.89
Airmass coefficient 1 (a1)	0.884 +/- 0.025
Airmass coefficient 2 (a2)	0.092 +/- 0.07
Scatter in the residuals of the lightcurve fit is	2.8 %
Best Comparison Star	#3 - [521, 325]
Optimal Method	PSF photometry
Transit Duration (day)	0.0863 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.094 ± 0.045 vs 0.1368 (deviation 0.95σ)
Duration fit vs published	0.0863 d vs 0.0758 d (deviation 1.08σ)
Mid-time O–C	15.8 min
Depth SNR	2.1
Residual scatter	2.8 %

Light curve

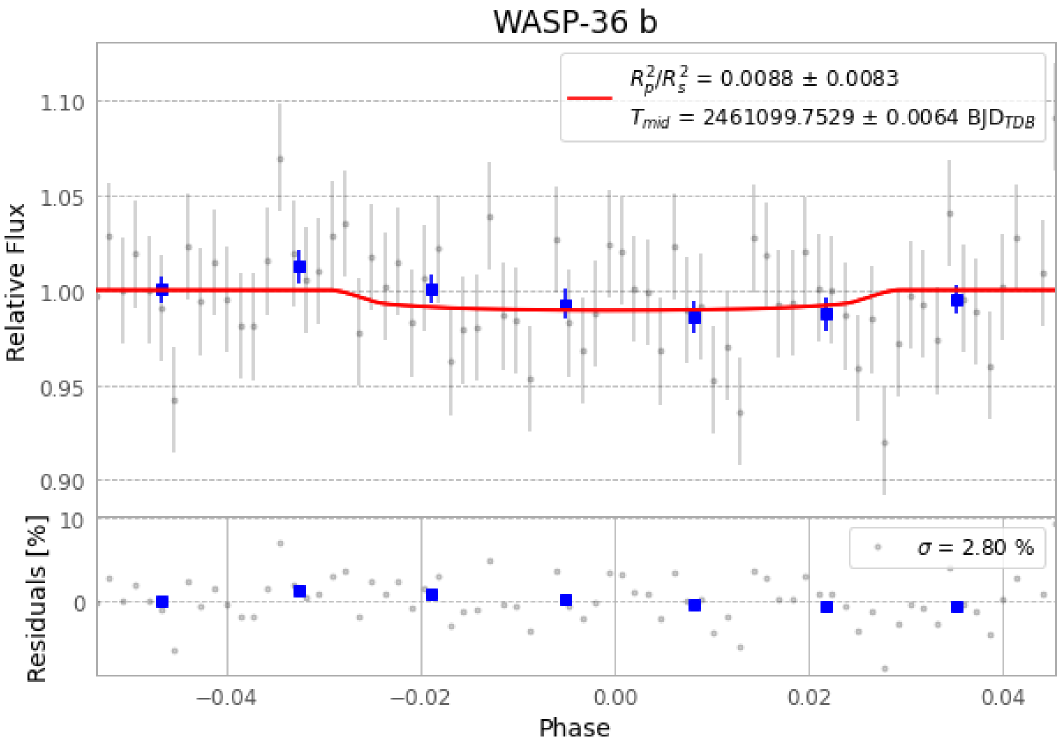

Figure 1 — FinalLightCurve_WASP-36 b_2026-02-27.png (SHA-256 8045fbcd1ad90bb7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 153], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cdd1e8ae7534b4ad…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f6a95f5`

Data provenance

74 FITS frames (49 MB), `WASP-36260228035716.FITS` → `WASP-36260228073616.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260228.log (c44690f37f13…), FinalLightCurve_WASP-36 b_2026-02-27.png (8045fbcd1ad9…), FinalLightCurve_WASP-36 b_2026-02-27.csv (01c012d4b84c…)

Compute resources

Wall time 185.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-27 18:17Z.